



Ion uptake and pairing in membranes: The pore model

Viatcheslav Freger 

Wolfson Department of Chemical Engineering, Technion - Israel Institute of Technology, Haifa, 32000, Israel

Grand Water Research Institute, Technion - Israel Institute of Technology, Haifa, 32000, Israel

Grand Technion Energy Program, Technion - Israel Institute of Technology, Haifa, 32000, Israel

ARTICLE INFO

Keywords:

Ion-pairing in membranes

Ion uptake

Transport and conductance

Pore and primitive models

Association constant

Dielectric exclusion

ABSTRACT

Ion uptake plays a critical role in membrane separations used for water purification and electro-membrane processes. Although models of ion uptake usually consider free dissociated ions, ion-pairing was suggested to play an important role as well and but its understanding in the context of membrane transport is still insufficient. The paper systematically develops the pore model of ion-pairing in membranes, which considers the membrane as a micro-heterogeneous system, composed of interspersed water-rich domains (pores accommodating ions) and a low-dielectric matrix. It is shown that this picture significantly departs from the predictions of the primitive model of the membrane as a uniform dielectric, in which dielectric exclusion and pairing of ions are rigidly related thus pairing of mobile ions is always negligible. Within the pore model, the effect of the ion-pairing constant enhanced by the low-dielectric properties of the matrix may outweigh the opposite effect of dielectric exclusion thereby mobile ions may be present within the membrane mainly as ion-pairs rather than free ions. We discuss various implications of this result for salt and ion uptake, transport, and conductance in neutral and charged membranes and outline needs for further research towards predictive modeling.

1. Introduction

Ion uptake and transport play key role in many membrane separations, including reverse osmosis, nanofiltration, electrodialysis, membrane electrolysis, and fuel cells [1,2]. Classical models of these processes, such as the Donnan and Poisson–Boltzmann-based charged-nanopore models, mainly considered ions in membranes as free dissociated ions, in presumed analogy with aqueous solutions, where salts are usually strongly dissociated [1–5]. The dissociation is however known to be incomplete even in solutions. Pairing or association of ions has been introduced by Bjerrum with the purpose to resolve the inherent problems of the Debye–Hückel (DH) theory of ionic solutions [6–8]. Despite being an *ad hoc* concept, it proved a sound and useful approximation that could transparently address the limitations of the DH and other mean-field models. Its key feature is the view of ion-pairs of strong electrolytes as distinct species in equilibrium with free ions, similar to non-dissociated forms of genuine weak electrolytes such as carboxylic acids.

Ion-pairing effectively reduces the thermodynamic activity of ions and their contribution to electrical conductance, which has been used to quantify this phenomenon, especially, in organic solutions [6,9]. For common inorganic ions in aqueous solutions, the high dielectric constant of water weakens ion-ion interactions thus the entropy gain, of the order $k_B T$, dominates, which is why salts tends to dissociate

to free ions. The degree of ion-pairing in water is then significant mainly for multivalent ions such as Ca^{2+} , Mg^{2+} , and SO_4^{2-} at high concentrations, but it is fairly small for common monovalent ions such as Na^+ , K^+ and Cl^- [9]. The situation, however, changes drastically in non-polar solvents, where substantially lower dielectric constants enhance ion-ion interactions and promotes formation for ion-pairs and larger associates [6,9]. This is the main reason behind low solubility of inorganic salts in organic solvents.

Most synthetic membranes used in desalination, electrodialysis, and fuel cells are made of organic polymers and share common features with organic media. However, they differ from organic liquids in two main aspects. First, they often contain fixed charges and an equivalent amount of mobile counter-ion [1,10], enforcing a high ion concentration on a medium, in which ion solubility is inherently low. Fixed charges in such membranes are then particularly prone to ion-pairing, affecting ion exclusion, diffusion, and conductance [11]. Second, to perform as selective barriers or adsorbents, membranes are usually rendered insoluble via crosslinking (resins) or a large content of hydrophobic units (ionomers) [1,10]. When swollen in water, both resins and ionomers are prone to forming a micro-heterogeneous structure, in which water-rich inter-polymer domains, reminiscent of water-filled molecular pores, are segregated from non-polar water-lean matrix, as illustrated in Fig. 1A [12–16].

E-mail address: vfreger@technion.ac.il.

<https://doi.org/10.1016/j.memsci.2025.123795>

Received 7 December 2024; Received in revised form 28 January 2025; Accepted 30 January 2025

Available online 7 February 2025

0376-7388/© 2025 The Author. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

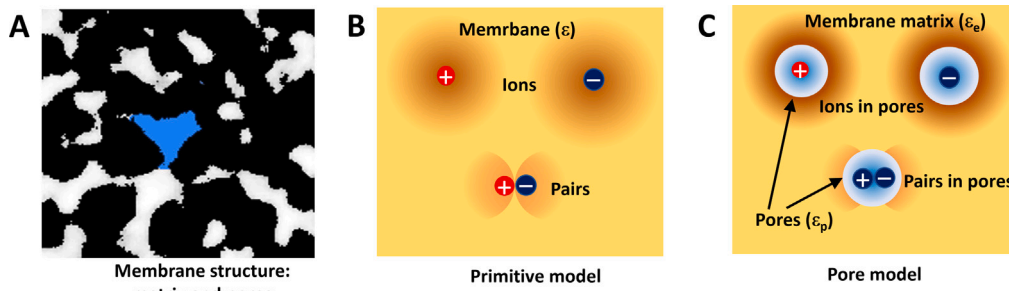


Fig. 1. A schematic picture of the actual membrane structure and the two discussed models. (A) Two-phase membrane microstructure: matrix is black, pores are white, colored in blue is a single pore. (B) The primitive model (PM) of a membrane: ions are charged balls and the membrane is a uniform continuum of dielectric constant ϵ equal to the effective average of the entire membrane phase. (C) The pore model: water-filled pores of dielectric constant ϵ_p dispersed in a low-dielectric matrix; the dielectric constant of the medium surrounding the pore is the effective average of the entire membrane phase ϵ_e . Ions are charged hard-spheres present within the pores. In panels (B) and (C), the color shades depict the polarization energy density around ions and pairs.

The above points and importance of ion-pairing in general were recognized to various extent and ion-pairing was included in models of transport and partitioning already in early days of membrane science [17–19] and also more recently [20–23]. To the best of the authors knowledge, with perhaps only one exception [19], most of these studies modeled pairing by viewing the membrane as a dielectric continuum or computing pairing in the pores based on association constant in solution. As shown below, both such assumptions miscalculate the strength of pairing and its relative role, compared with free ions, in ion uptake and transport in membranes. The present paper focuses on understanding how micro-heterogeneity affects ion-pairing, towards the ultimate goal of getting closer to predictive modeling.

2. Theory

2.1. Addressing ion-pairing in modeling

Before considerin the physical picture in detail, it is expedient to outline how, in general, models should incorporate ion-pairing. For clarity, we address here only ion uptake and leave out permeation to avoid complications related to the effect of pairing on ion mobility, discussed in Section 3.2. The key point is that, unlike genuine molecules, ion-pairs do not “partition” between the solution and membrane phases as a distinct species. Rather, they keep breaking up and reforming dynamically from free ions in each phase. Since the dynamics is reversible and very fast, the uptake of ions, as interrelated uptake of free ions and pairs, needs to include the following equilibrium relations:

(i) Partitioning of free ions between solution and membranes based on the difference in (free) ion activity coefficients between the phases. This is ultimately expressed by the salt-injection coefficient S_0 defined below;

(ii) Equilibrium between free ions and all possible types of ion pairs in the membrane, including pairs formed by mobile ions (mobile pairs) and by fixed and mobile ions (fixed pairs). For each type of pair, the equilibrium is expressed by means of a respective pairing constant K_m .

(iii) electroneutrality of the membrane phase that requires the zero total charge of unpaired fixed charged, free mobile ions, and all types of pairs.

Notably, for multivalent ions, pairs may be charged and the resulting relations for charged membranes vary significantly, depending on the type of the salt and pairs included in the model. For different types of single salts, they were fully developed in a previous publication [24], viewing S_0 and K_m ’s as parameters. Here, for illustration, consider the simplest case of a 1:1 salt, such as NaCl, in a non-charged membrane. The total measured salt concentration in the membrane, identical for cations and anions, will sum up the free salt and ion-pairs concentrations, $C_m = S_0 C_s$ and $C_{pair} = K_m C_m^2$, respectively, and depend on the salt concentration in solution C_s as follows

$$C_m + C_{pair} = S_0 C_s + K_m (S_0 C_s)^2. \quad (1)$$

This relation shows that the contribution of pairs to salt uptake will increase with C_s and the dependence of salt uptake on salt concentration may change from linear to quadratic between low and high concentration ranges. This was indeed observed in early studies of weakly charged cellulose acetate membranes [17–19].

Similarly, for charged membranes and high salt concentrations, pairing may affect the trend of salt uptake due to enhanced salt uptake as ion-pairs and Donnan exclusion weakened by reduced fixed charge [25,26]. However, this requires that intra-membrane environment enhances K_m more significantly than it suppresses S_0 (cf. Eq. (1)).

Note Eq. (1) does not assume that ions form pairs in solution, emphasizing that ion-pairing in the membrane is not “partitioning” of ion-pairs. If salts do form ion-pairs in solution, their solution concentration is $K_s C_s^2$, where K_s is the pairing constant in solution. Thus the ratio of ion-pair concentrations in the membrane and solution will be $(K_m/K_s)S_0^2$, yet we reiterate it is a not a genuine partitioning coefficient, but a result of several equilibrium relations.

Finally, we note that full modeling of ion uptake will require that K_m and S_0 be known for each pair of ions, including fixed charges. Likely, the available experimental data may not allow a straightforward evaluation of these parameters [24]. This is where the approach developed in this paper may help get closer to predicting them *a priori* using a few known physical characteristics of the ions and the membrane.

2.2. What controversy this study aims to resolve

Usually, liquid solutions of salts are fairly homogeneous down to single-molecule scale, which suggests they may be reasonably modeled using the so-called primitive model (PM) that views the solvent as a dielectric continuum and ions as charged hard spheres (Fig. 1B) [27, 28]. A subtle point highlighted recently is that, within PM, there is a strong inherent connection or *coupling between ion solubility and ion-pairing*, whereby low-solubility inevitably precludes substantial pairing, as outlined next [11].

PM relates both solubility (solvation) and pairing to the dielectric constant of the medium, ϵ , conveniently represented by the Bjerrum length λ , defined as the distance, at which the energy of (monovalent) ion-ion interaction U equals the thermal energy $k_B T$ [8], i.e.,

$$\lambda = \frac{e^2}{4\pi\epsilon_0\epsilon k_B T}, \quad (2)$$

where e is the electron charge and ϵ_0 is the permittivity of vacuum. On the other hand, the maximum absolute value of U upon pairing, U_{pair} , is reached when two ions come in contact, to the minimal inter-ion distance b , equal to the sum of their radii. The ratio between $k_B T$ and U_{pair} or, equivalently, the two above distances defines the reduced temperature T^* [27], as follows

$$T^* = \frac{k_B T}{U_{pair}} = \frac{b}{\lambda}. \quad (3)$$

Its magnitude is what differentiates between high-dielectric media, $T^* \sim 1$, such as aqueous solutions, and low-dielectric ones, $T^* \ll 1$, to which most synthetic membranes belong and which are in the focus of the present study [11]. The Born equation gives the solvation or self-energy energy of free ions of radius $r_i = b/2$ and charge z , which is the energy of dielectric polarization of the medium surrounding the ion (cf. Fig. 1B) [29–31]. This energy, W , and corresponding ion activity coefficient, γ , are

$$\ln \gamma = \frac{W}{k_B T} = z^2 \frac{\lambda}{b} = z^2 \frac{1}{T^*}. \quad (4)$$

Conversly, Bjerrum's pairing constant derives from the ion-ion interactions, which at sufficiently short distances involve mainly direct Coulomb interaction between two ions thus $U(x) = -z_+ z_- \lambda / x$, where z_+ and z_- are the ion charges and x is the distance between the ions [6, 7, 32]. Based in the Boltzmann factor $\exp[-U(x)/k_B T]$ and spherical symmetry, the following equation sums up the probabilities of finding the two ions in the paired state and yields the pairing constant that connects the concentrations of pairs, C_{pair} , and free monovalent ions, C_+ and C_- , in the membrane, as follows

$$K_m = \frac{C_{pair}}{C_+ C_-} = \int_b^q \exp\left(\frac{z_+ z_- \lambda}{x}\right) 4\pi x^2 dx \simeq 4\pi b^3 \frac{T^*}{z_+ z_-} \exp\left(\frac{z_+ z_-}{T^*}\right), \quad (5)$$

where $q = z_+ z_- \lambda / 2$ is Bjerrum's cut-off distance that sets apart the paired and free-ion states [6, 32].

The key point is that above PM relations, Eqs. (4) and (5), equivalent to most models of ion-pairing in membranes proposed both in early studies [17, 18] and more recently [11, 20–22], depend on the same parameters and predict that a *non-charged* low- T^* membrane will *always contain a negligible amount of ion-pairs compared with free ions* [11]. Indeed, taking a salt of monovalent ions as an example, both C_+ and C_- equal the free salt concentration in the membrane C_m . In turn, C_m may not exceed the salt solubility in the membrane, whose upper limit is $C_{max}/\gamma = C_{max} e^{-1/T^*}$ where $C_{max} \simeq b^{-3}$ is the concentration of densely packed non-interacting ions (i.e., solid salt), as the hypothetical reference state in equilibrium with solution. Adding in this reference state attractive ion-ion interactions will only reduce this upper limit for C_m and, if the membrane is in equilibrium with a solution below solubility, the free-ion concentration in the membrane will be reduced even further. We can then find the maximal fraction of pairs, as follows

$$\frac{C_{pair}}{C_m} = \frac{K_m C_+ C_-}{C_m} \simeq K_m C_m < b^3 T^* e^{1/T^*} \times C_{max} e^{-1/T^*} \simeq T^*. \quad (6)$$

This shows that, for $T^* \ll 1$, this fraction in the membrane is always expected to be small. This means the effect of membrane's low-dielectric properties on ion exclusion, reducing the free ion concentration in the membrane, outweighs their effect on pairing constant. Obviously, being a minor species, ion-pairs must also negligibly contribute to salt or ion transport. However, the aforementioned studies of cellulose acetate membranes indicate the opposite [17–19].

Similar but somewhat lengthier analysis may be applied to charged membrane. It shows that, for $T^* \ll 1$, the fixed charges will form pairs with counter-ions thus “effective” (dissociated) membrane charge will be significantly reduced and co-ion exclusion weakened. However, the dielectric ion exclusion will still keep the co-ion concentration smaller than the effective fixed charge, thus the Donnan mechanism will still operate. Also, similar to the neutral membranes, pairing of free mobile ions, i.e., co- and counter-ions, will be negligible and contribute insignificantly to salt uptake. The partitioning of free salt should then follow the trend of the Donnan exclusion, only quantitatively but not qualitatively modified by other exclusion mechanisms [11]. In particular, the salt uptake in a charged membrane should always increase with salt concentration and so will the salt permeability.

A weaker than expected but still increasing trend was indeed observed for monovalent ions, both in ion-exchange and nanofiltration membranes [33, 34]. However, in nanofiltration membranes, salts of divalent Ca^{2+} and Mg^{2+} ions display an opposite trend of permeability decreasing with salt concentration [34–36]. This is illustrated in Fig. 2,

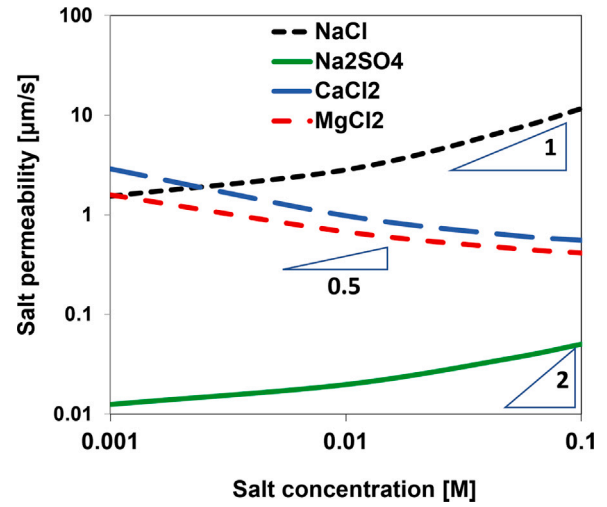


Fig. 2. Variation of salt permeabilities with salt concentration for NF270 nanofiltration membrane and single salts of different mono- and divalent ions. Permeability variation was deduced by fitting to filtration data for different applied pressure and feed solution concentrations. Adopted from reference [X]. The triangles indicate the theoretical slopes predicted by the Donnan model.

showing variation of salt permeabilities with feed salt concentration observed in nanofiltration of different salts, strongly deviating from the Donnan-based trends. This rules out permeation controlled by free ions only and suggests that pairing, much stronger than the PM analysis predicts, may be involved. This conclusion and, specifically, the strong reduction of fixed charge by pairing (binding) of counter-ions to fixed reverse osmosis and nanofiltration membranes [25, 26, 37].

Evidence of strong ion association was also obtained for ion-exchange membranes, manifested in much lower than expected ion conductivity and osmotic pressure within membranes [38, 39]. Recently, it was specifically concluded that divalent SO_4^{2-} co-ions are present within such membranes as monovalent pairs such as HSO_4^- or NaSO_4^- . The conclusion was based in the observed concentration dependence of sulfate partitioning in negatively charged ion-exchange membranes, strongly deviating from the trend expected for divalent co-ions, but closely matching that for monovalent ones [24, 40].

Despite much evidence, the physics behind excessive pairing of mobile ions in membranes, contradicting the PM predictions, has remained elusive so far. In this paper, we will demonstrate that such behavior is, in fact, expected, once we depart from the PM picture and consider micro-heterogeneity in the form of microscopic water-filled dynamic “pores” or water- and ion-rich domains, dispersed in a relatively non-polar matrix, as illustrated in Fig. 1C. Recently, this picture was successfully used to describe dielectric exclusion of ions in membranes, offering significant improvement over PM (Eq. ((3))) [31, 41]. The present paper utilizes the same physical picture - the pore model - to model ion pairing.

2.3. The pore model: dielectric exclusion of free ions

Before proceeding to pairing, let us briefly recap the pore model results for dielectric exclusion [31]. In general, the strength of exclusion is affected by the pore shape and location of the ions within the pore [3, 29, 30]. The complete model then needs to average over pore geometries found in the membranes and possible ion locations. Yet, we will view the pores as spherical and consider ions or pairs located at their center. As discussed previously [31], the approximation of spherical pores is reasonable when the pores and matrix are interspersed randomly and lack macro-scale anisotropy, which is most common in polymeric membranes. Placing ions at the center is

a sensible approximation too, since this position corresponds to the minimum energy and thereby has the largest statistical weight of all possible states. Obviously, this somewhat overestimates the energy gain upon ion uptake but the required correction is fairly small [19,29]. With these reservations in mind, the picture offers maximal simplicity and clarity and presumably captures most essential physics. In addition, we will also assume pores are small enough to contain at most one free ion not counter-balanced by an oppositely charged one. That still permits presence of other ions in the pore, provided their total charge is zero and the pore content is water-like, i.e., behaves as a high dielectric.

The partitioning of *free ions* from a solution (s) to a membrane pore (p), surrounded by an effective medium (e) composed of the membrane matrix and all other pores (Fig. 1C), will be equal to the ratio of respective activity coefficients, γ^s/γ^p . Assuming they are dominated by solvation (dielectric exclusion), this ratio will be determined by the difference in the respective Born self-energies, $\Delta W = W_p - W_s$, as follows [31]

$$\ln\left(\frac{\gamma^s}{\gamma^p}\right) = -\frac{\Delta W}{k_B T} = -\frac{z^2}{2} \left(\frac{\lambda_e - \lambda_p}{r_p} + \frac{\lambda_p - \lambda_s}{r_i} \right), \quad (7)$$

where λ 's are the Bjerrum lengths of respective media, and r_p and r_i are the pore and ion radii.

Notably, Eq. (7) may not be used directly to calculate ion partitioning coefficient S , which would require adding to Eq. (6) a term accounting for a *a priori* unknown inter-phase Donnan potential ϕ_D . However, ϕ_D is canceled out when one considers neutral combinations of ions. For a $M_{z_-}A_{z_+}$ salt composed of z_- cations M^{z_-} and z_+ anions A^{z_-} , ϕ_D will be canceled out in the free-ion concentration product $C_{+}^{z_-}C_{-}^{z_+}$. The *salt injection coefficient* S_0 [11] is then defined based on the ratio of such products in the pores and in the solution normalized per ion, as follows

$$\ln S_0 = \frac{1}{z_+ + z_-} \ln \left(\frac{C_{+}^{z_-}C_{-}^{z_+}}{C_{+}^{z_-}C_{-}^{z_+}} \right)_p = \frac{1}{z_+ + z_-} \ln \left[\left(\frac{\gamma_{+}^{z_-}}{\gamma_{+}^{z_-}} \right)^{z_-} \left(\frac{\gamma_{-}^{z_+}}{\gamma_{-}^{z_+}} \right)^{z_+} \right]. \quad (8)$$

In the case of a non-charged membrane, S_0 is identical with the partitioning coefficient of the fully dissociated (free) salt between the membrane and solution. As such, it expresses the affinity of the membrane to uptake of the salt as *free ions*. In a charged membrane, S_0 needs to combine with the electroneutrality and association equilibria to compute free-ion concentrations [24] (see Section 2.1). Combining Eqs. (7) and (8) yields

$$\ln S_0 = -\frac{z_- \Delta W_{+} + z_+ \Delta W_{-}}{(z_+ + z_-)k_B T} = -\frac{z_+ z_-}{2} \left[\frac{\lambda_e - \lambda_p}{r_p} + \frac{\lambda_p - \lambda_s}{z_+ + z_-} \left(\frac{z_+}{r_+} + \frac{z_-}{r_-} \right) \right]. \quad (9)$$

In most membranes, $\lambda_e \gg \lambda_p$ and, on the other hand, the pore and ion radii, r_p and r_i , usually do not differ as much and $\lambda_p - \lambda_s$ is small. This suggests that the second terms in brackets in Eqs. (7) and (9), dependent on the ion sizes, is fairly small too. Therefore, as a first approximation, we can ignore it and simplify Eqs. (7) and (9) for the dielectric energy of the ion and for the salt injection coefficient S_0 by keeping the first term only, as follows

$$\ln\left(\frac{\gamma^s}{\gamma^p}\right) = -\frac{\Delta W}{k_B T} \approx -\frac{z^2}{2} A, \quad (10)$$

$$\ln S_0 \approx -\frac{z_+ z_-}{2} A, \quad (11)$$

where

$$A = \frac{\lambda_e - \lambda_p}{r_p} \quad (12)$$

is a key dimensionless parameter of the membrane defining the strength of dielectric exclusion in its pores. In membranes, λ_e is of the order of 5–10 nm and much larger than λ_p and r_p , therefore, $A \gg 1$. Comparison of Eqs. (4) and (10) suggests that $2/A$ may be seen as a modified T^* of the pore model, replacing the T^* of PM defined by Eq. (3). Since S_0 is exponentially related to A , a small A is what makes the dielectric exclusion in membranes strong.

2.4. The pore model: relations for ion-pairing constant

In most membranes, the cut-off distance $q = z_+ z_- \lambda_s / 2 \sim (z_+ z_-) 0.35$ nm that sets ion-pairs apart from free ions is not much below the characteristic pore radius and may even be above it. It may then be reasonable to re-define a pair as two oppositely charged ions in the same pore. The ion-pairing then becomes equivalent to the transition from the situation when two single ions, initially in different pores (free ions), move to the same pore (ion-pair), as illustrated in Fig. 3A. Computation of the pairing constant K_m , the main parameter of interest, needs to integrate over the possible states of the pair weighed with their energy, as in Eq. (5).

Intuitively, we could estimate the pairing energy by considering two contributions: the change of the self-energy upon pairing (ion-solvent interaction), ΔW_{pair} , and the energy of ion-ion interaction, $U(x)$. The former should account for the fact that the pair has a smaller charge $z_{pair} = z_+ - z_-$ and polarizes the surrounding medium less than two separate free ions, as schematically indicated in Figs. 1B and C and 3A by color shades representing polarization energy density around ions and pairs. Note that this ΔW_{pair} contribution is absent in PM treatment of association (Eq. (5) and Fig. 1B). Using approximate Eq. (10) for both ions and the pair, the self-energy would change upon pairing by

$$\frac{\Delta W_{pair}}{k_B T} \approx \frac{A}{2} [(z_+ - z_-)^2 - z_+^2 - z_-^2] = -z_+ z_- A. \quad (13)$$

The second contribution, ion-ion interaction energy $U(x)$, varies with the distance x between the ions and their position within the pore. To calculate it approximately, we simplify by placing one ion, say z_- , at the center of a pore and consider the electric potential $\phi(x)$ induced by this ion at distance x , thus $U(x) = z_+ e \phi(x)$. By setting $\phi(\infty) = 0$ and using elementary electrostatics, we obtain for $x \leq r_p$

$$\frac{U(x)}{k_B T} \approx -z_+ z_- \left(\frac{\lambda_e - \lambda_p}{r_p} + \frac{\lambda_p}{x} \right) = -z_+ z_- \left(A + \frac{\lambda_p}{x} \right). \quad (14)$$

The sum of Eqs. (13) and (14) should approximate the total energy gain upon pairing. However, a more rigorous analysis shows that, within the same-order approximation, such formal summation of Eqs. (13) and (14) would double-count the change of polarization energy expressed by the term $-z_1 z_2 A$ (e.g., see problem 3-30 in [42]). With this correction, the energy of pairing is given by Eq. (14) only. As in Eq. (5), the corresponding Boltzmann factor may be used to compute the pairing constant K by statistical integration over all paired states, $b \leq x \leq r_p$, as follows

$$K_m = \int_b^{r_p} \exp\left(-\frac{U(x)}{k_B T}\right) 4\pi x^2 dx \approx e^{z_- z_+ A} \int_b^{r_p} \exp\left(-\frac{z_+ z_- \lambda_p}{x}\right) 4\pi x^2 dx. \quad (15)$$

Inspection of Eq. (15) shows that, if within pores $\lambda_p \approx \lambda_s$, the integral in the last expression approximately represents the pairing constant for the same ions in solution K_s , except that the upper limit is r_p instead of q . The prefactor $\exp(z_- z_+ A)$ that multiplies the integral then expresses the enhancement of association in pores, relative to solution, which is a part of the “partitioning coefficient” of ion-pairs (Section 2.1), exponentially dependent on the same parameter A that controls the dielectric exclusion.

To approximate how the last integral depends on r_p and deviates from K_s , since r_p differs from q , we may expand the integrand $4\pi x^2 \exp(2q/x)$ around $x = q$ and integrate between q and r_p which ultimately yields the following 2nd order approximation

$$K_m \approx e^{z_- z_+ A} \left(K_s + 4\pi e^2 \left[q^2 (r_p - q) + \frac{1}{3} (r_p - q)^3 \right] \right). \quad (16)$$

Fig. 4 illustrates that the approximate expression in brackets works reasonably up until $r_p \sim 2q$ for representative values of b , λ_p , and r_p . Note that K_s is known for many inorganic ions that associate noticeably in solution, i.e., when K_s is substantial [9]. Presumably, the experimental K_{sol} may incorporate short-range water structure effects,

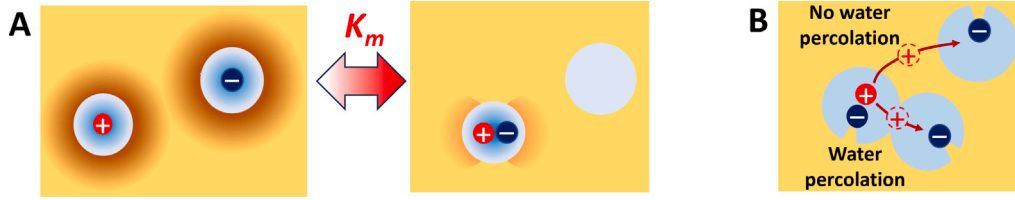


Fig. 3. (A) A sketch of the free ions/ion-pair equilibrium in the pore model. Free dissociated ions initially found in different pores reduce the total polarization energy by moving to the same pore to form an ion-pairs. K_m is the pairing constant. Color shades indicate the polarization energy density. (B) Schematic illustration of mobile counter-ion (red) hopping between fixed charges (blue) in different pores with transient dissociation of the counter-ion/fixed charge pair in the scenarios when the water microphase does and does not percolate.

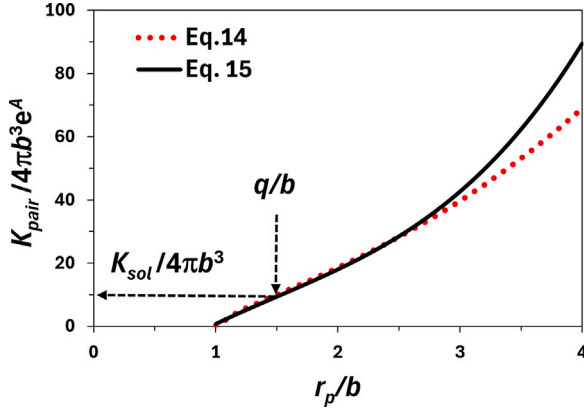


Fig. 4. The variation of the pairing constant K_{pair} , defined by Eq. (14), with the pore size r_p , based on numerical integration and approximation given by Eq. (15). r_p and K_{pair} are made dimensionless by scaling with the ion-pair size b and with $4\pi b^3 e^A$, respectively. The parameters used in calculations are $b = 0.3$ nm, $\lambda_p = 0.9$ nm, $z_+ = z_- = 1$. The value of K_{sol} scaled with $4\pi b^3$, corresponding to $r_p = q = \lambda_p/2$ where Eqs. (14) and (15) match exactly, is indicated.

ignored in the present model. Thus, in Eq. (16), it may be expedient to use and correct the experimental K_s , rather than compute it by integration using $x = q$ as the upper limit. Also note that the term after the prefactor $\exp(z_- z_+ A)$, i.e., the last integral in Eq. (15), must always exceed the pore volume $V_p = \frac{4\pi}{3} r_p^3$ (minus the pair volume). This mean $V_p \exp(z_- z_+ A)$ represents the lower bound for K_m , corresponding to weak ion-ion interactions, which will be used below.

2.5. Ion-pair versus free-ion partitioning

Previously, we saw that PM treatment predicts that ion-pairs always contribute negligibly to partitioning and transport of mobile ions, except for association of counter-ions with fixed charges. Let us check whether the same holds true for the pore model. Consider partitioning of a $M_{z_-} A_{z_+}$ salt, present in solution at concentration C_s , to a non-charged membrane. Neglecting association in solution, free-ion concentration in the membrane will be

$$\frac{(C_+)_m}{z_-} = \frac{(C_-)_m}{z_+} = S_0 C_s \approx C_s \exp\left(\frac{z_+ z_-}{2} A\right). \quad (17)$$

Since the pairing constant is bound from below by V_p , the minimal amount of pairs in the membrane relative to concentration of free ions (paired fraction) may be estimated as follows (cf. Eq. (6))

$$\frac{C_{pair}}{C_m} \approx \frac{K_m (S_0 C_s)^2}{S_0 C_s} = K_m S_0 C_s \approx C_s V_p \exp\left(\frac{z_+ z_-}{2} A\right). \quad (18)$$

In stark contrast with Eq. (6), in the last expression, the exponential term of K_m is not canceled out entirely and only partly offset by that of S_0 . As a result, in the pore model, dielectric exclusion (solvation) and pairing are *no more coupled* as rigidly as in PM, and *strong ion exclusion no more precludes strong association* within the membrane.

For example, consider representative values $c_s = 0.1$ M, $r_p = 0.3$ nm, $\lambda_e = 5$ nm ($\epsilon_e = 10$), and $\lambda_p = 1$ nm, slightly above $\lambda_s = 0.7$ nm. This corresponds to $C_s V_p \approx 10^{-2}$ and $A \approx 13.3$. It is easily seen that, even for $z_+ = z_- = 1$ and this moderate C_s , the large exponential factor $\exp(A/2) \approx 10^3$ in Eq. (18) will outweigh the $C_s V_p$. This indicates that the mobile ions in the membrane will tend to form pairs in the present sense, with only relatively few ions remaining free at this and higher salt concentrations. The salt uptake will then mainly occur as uptake of pairs rather than free ions, with many consequences for salt transport and conductance (see Discussion). Obviously, this effect will become stronger for more charged ions and at higher concentrations. However, the pore size effect may be more complex, since r_p enters both V_p and A in Eq. (18). The dependence of $K_m S_0$, hence the paired fraction, on r_p is then not monotonic, showing a minimum at $r_{min} = \frac{1}{6} z_+ z_- (\lambda_e - \lambda_p)$. For the above numbers, this corresponds to $r_{min} = 0.7$ nm, below which the degree of pairing will increase in smaller pores.

3. Discussion

3.1. Why the pore model differs from PM

How can we understand the drastic difference between the pore model and PM predictions? Essentially, it boils down to the fact that, in PM, pairing offsets only a fraction of the ion self-energy, e.g., exactly one half for equally sized ions forming a neutral pair. The remaining half is mostly contained in the space between and very near the ions, where the field of the ions is not cancelled and remains strong. In contrast, in the pore model, this residual energy is greatly reduced by high-dielectric environment within the pore. Thus the long-range polarization energy, substantial for free ions, vanishes or gets strongly reduced when oppositely charged ions enter the same pore.

Curiously, a similar situation arises when ions enter a narrow space enclosed by a conductor, such as a single-wall carbon nanotube (CNT). When a single ion enters the CNT interior, CNT walls acquire and delocalize the ion charge and polarize the surrounding medium. However, when two ions form a pair within a CNT, the charge vanishes and so does the polarization energy, which turns out to be the dominant factor that promotes ion-pairing and ion uptake within CNT [43]. In the present case, the high polarizability of the water filling the pore makes it somewhat similar to a conductor. Nevertheless, the equivalence is not total, as the polarizing field around a neutral pair a conducting pore totally vanishes but is still finite around a dielectric pore. We can, however, estimate the magnitude of the residual polarization energy of such a pair, with one ion at the center of the pore and the second at a distance b from it, as that of a dipole with a dipole moment $\mu = eb$ relative to the pore center using the methods developed for analysis of molecular polarization in dielectrics [44]. The result is

$$\frac{W_{dipole}}{k_B T} = -\frac{\epsilon_e - \epsilon_p}{2\epsilon_e + \epsilon_p} \frac{\mu^2}{4\pi\epsilon_0\epsilon_p k_B T r_p^3} \approx \frac{\lambda_p b^2}{r_p^3}. \quad (19)$$

The last approximate expression follows from $\epsilon_e \ll \epsilon_p$. Using representative values $b = 0.3$ nm, $\lambda_p = 1$ nm, and $r_p = 0.3$ nm, we estimate $W_{dipole} \approx 2k_B T$. This energy is finite and positive, i.e., it is a

penalty that discourages pairing and, for completeness, should be added to $U(x)$ in Eq. (15). However, note the last expression in Eq. (19) may be presented as the product of λ_p/r_p , which should be much smaller than $A \simeq \lambda_e/r_p$ (Eq. (12)), and $(b/r_p)^2$, which is less than 1, as long as the pore can accommodate a pair. Therefore, W_{dipole} should be always small compared with $U(x)$, whose absolute value is at least as large as $Ak_B T$ (see Eq. (14)). Moreover, Eq. (19) overestimates this correction, since the assumed excentric position of the dipole does not correspond to the minimal energy. Thus it may be ignored as a first approximation.

However, very small pores, such as ~ 0.2 nm pores of reverse osmosis membranes [45], may be unable to accommodate pairs. The pairing and pair uptake may then be strongly suppressed by steric exclusion while smaller free ions may still access the pores, in which case the present picture needs to be modified significantly. This point warrants further investigation in future research.

3.2. Implications for ion uptake in charged membranes

The extension of above analysis to charged membrane is fairly straightforward and will only need to combine the expressions for K_m and S_0 with equilibrium relations reported previously for different salts [24]. Such full analysis is not pursued here, but it is expedient to mention briefly a few key points.

In agreement with actual observation and in contrast with PM predictions (Section 2.2), the pore model does allow K_m to be large enough, compared with S_0 , to allow both effective elimination of the fixed charge via pairing with counter-ions and significant uptake of salt as mobile ion-pairs formed by two mobile ions (Section 2.4). For multivalent counter-ions, such as Ca^{2+} , the fixed charge may even reverse, which is also precluded in PM but possible in the pore model. Once the free salt uptake exceeds the effective fixed charge, the Donnan exclusion will become negligible, and the membrane will exclude free salt essentially as a neutral membrane, i.e., using only non-Donnan mechanisms such as dielectric and steric exclusion. The salt (co-ion) uptake and permeability will then depend about linearly on solution concentration, as opposed to a stronger, non-linear dependence characteristic of the Donnan exclusion. In case when salt partitions to the membrane predominantly as mobile ion-pairs, the partitioning will increase non-linearly.

Interestingly, the resulting non-linear trend of ion-pair uptake may bear superficial similarity to the Donnan exclusion. For instance, for monovalent ions in a neutral membrane, concentration of mobile pairs in the membrane will be $C_{pair} = K_m(S_0 C_s)^2$ (see Eq. (1)), i.e., will have a square dependence on C_s . The Donnan model predicts essentially the same dependence, except that the term $(S_0 C_s)^2$ is multiplied by $1/X$, where X is the effective membrane charge, instead of K_m . Thus it may be difficult to tell from experimental data whether the observed trend reflects Donnan exclusion or pairing. Notably, since symmetric pairs are neutral and unaffected by the Donnan exclusion, this square dependence for C_{pair} will be valid for a monovalent salt regardless of whether the membrane is neutral or charged. Thereby, co-ion uptake may simultaneously involve both free co-ions, obeying Donnan exclusion, and ion-pairs, which may be hard to differentiate based on salt uptake data only. The situation for multivalent ions forming asymmetric pairs may be even more complex [24,40]. Future studies will need to address these challenges.

3.3. Implications for ion transport

Ion pairing modifies ion uptake and thus directly affects the ion permeation and conductance in membranes through the ion partitioning factor. Yet, we emphasize here that pairing will strongly impact not just the partitioning of the co- and counter-ions but their mobility as well, as ions in pairs will be less mobile than free ions. The effect should be fairly moderate for mobile pairs, which will have a reduced but still substantial mobility. Thus, when ions may form only mobile pairs, as

all ions in non-charged membranes or co-ions in charged membranes, the effect of pairings on their mobility should be moderate.

However, pairing or binding of counter-ions with fixed charges should reduce their mobility more strongly. Their diffusion will then occur as hopping between fixed charges and require transient dissociation of the fixed pairs. The higher energy of the transient dissociated states will impose a barrier that will slow down the counter-ion diffusion. We showed previously that, within PM, the resulting mobility of counter-ions will be inversely related to the pairing constant K_m [11]. Measuring conductivity of charge membranes offers a straightforward way to quantify this effect. Since the counter-ion content in a salt-free membrane is fixed, the conductivity directly correlates with the macroscopic counter-ion mobility.

Specifically, increasing charge of counter-ions should exponentially affect K_m (cf Eqs. (5) and (15)) and reduce the mobility and conductivity accordingly. This is exactly what was reported recently, though the authors rationalized the results based on the Manning model of counter-ion diffusion [46]. This model relates the slowdown of diffusion to the variation of the electric potential within the membranes, assuming it is described by the screened mean-field potential of the Debye-Hückel type around fixed charges arranged in a cubic lattice [47]. Its mean-field nature makes it more appropriate for dilute polyelectrolytes in a high- T^* environment, which is principally different from the unscreened Bjerrum-type potential around individual fixed charges considered here. Also note that this diffusion model should not be confused with the Manning counter-ion condensation model [48], popular nowadays for analysis of ion uptake in ion-exchange membranes [40]. The latter model, alternative to the present one, is also based on a mean-field potential but fixed charges are assumed to form a uniformly charged rod embedded in a continuous solvent, which is principally different from the present picture too.

Within the pore model, the physical picture of the dissociation and hopping of counter-ions and corresponding reduction of mobility is more complex than in PM and may not show a simple inverse relation to K_m . Since the pore model no more views the membrane as a uniform medium, the rate of counter-ion hopping between the fixed charges or pores will depend on the pore connectivity and the need to cross a low-dielectric matrix as a transient state. Note that the thermodynamic picture of the pore model, Figs. 1C and 3A, and the calculation of K_m (Eq. (15)) do not consider any state of ion outside the pores. However, considering such states may be necessary for understanding mobility.

As schematically depicted in Fig. 3B, when the pore (water) volume fraction in the membrane is large enough for percolation of water-rich domains, the ions can indeed diffuse across the membrane without having to leave the water microphase. Such diffusion will still be affected by the geometric factors of porosity and tortuosity but, otherwise, could (inversely) correlate with the value of K_m given by Eqs. (15).

On the other hand, when the pore fraction drops below the percolation threshold, the need to cross the low-dielectric matrix may increase the barrier and suppress the mobility far more strongly. The consideration of water percolation as a prerequisite of good conductance are commonly invoked and reasonably explain the sharp drop of conductivity at reduced humidity in ion-conducting membranes, particularly important in fuel cell membranes [13,49,50]. We presume that, with appropriate modifications, the pore model may help compute the energy of different transient states, i.e., relevant energy barriers and their variation with hydration. This may clarify and quantify the relation between the membrane hydration and conductivity.

4. Conclusions

The pore model views the membrane as a micro-heterogeneous system, composed of interspersed water-rich pores, accommodating ions, and a low-dielectric matrix. The approximate relations, systematically derived in this study for the pore model, indicate that the low-dielectric property of the matrix may enhance ion-pairing more substantially than

the dielectric exclusion, opposite to the predictions of the primitive model of membrane as a uniform continuum. As a result, ions within the membrane may be present mainly as ion-pairs rather than free ions, which may have many implication for ion uptake, transport, and selectivity. In particular, in charged membranes, pairing could strongly reduce or entirely deactivate the fixed charge and suppress the mobility of counter-ions and membrane conductance. The resulting trends may vary for ions of different mobility and, on the other hand, be superficially similar to those predicted using classical Donnan model relations for free ions. This poses significant challenges for interpretation of the data and differentiation between mechanisms. Addressing these challenges and getting closer to a predictive theory may require both thorough modeling along the lines outlined here and elaborate experimental data on ion uptake and transport. This effort may also benefit from advanced tools such as molecular simulations and complementary characterization approaches such as dielectric spectroscopy [51].

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Viatcheslav Freger reports financial support was provided by Israel Science Foundation. Viatcheslav Freger reports financial support was provided by Israel Ministry of Innovation, Science and Technology (MOST) Israel, China-Israel Scientific Research Program.

Acknowledgments

The authors acknowledges financial support by Israel Science Foundation (grant 486/22) and by a grant No 0005658 “Novel Membranes for Highly-efficient Water Decontamination” from Israel Ministry of Innovation, Science and Technology (MOST) Israel within the China-Israel Scientific Research Program Israel-China program. The author also thanks Konstantin Morozov for expert advise and help with deriving relations of electrostatic and dielectric polarization.

Data availability

Data will be made available on request.

References

- [1] M. Mulder, M. Mulder, Membrane processes, Basic Princ. Membr. Technol. (1996) 280–415.
- [2] H. Strathmann, Ion-Exchange Membrane Separation Processes, vol. 9, Elsevier, 2004.
- [3] A. Yaroshchuk, M.L. Bruening, E. Zholkovskiy, Modelling nanofiltration of electrolyte solutions, Adv. Colloid Interface Sci. 268 (2019) 39–63.
- [4] R. Kingsbury, J. Wang, O. Coronell, Comparison of water and salt transport properties of ion exchange, reverse osmosis, and nanofiltration membranes for desalination and energy applications, J. Membr. Sci. 604 (2020) 117998.
- [5] P. Biesheuvel, S. Porada, M. Elimelech, J. Dykstra, Tutorial review of reverse osmosis and electrodialysis, J. Membr. Sci. 647 (2022) 120221.
- [6] R.A. Robinson, R.H. Stokes, Electrolyte Solutions, Courier Corporation, 2002.
- [7] J. Barthel, H. Krienke, W. Kunz, W. Kunz, Physical Chemistry of Electrolyte Solutions: Modern Aspects, vol. 5, Springer Science & Business Media, 1998.
- [8] A.Y. Grosberg, T. Nguyen, B. Shklovskii, Colloquium: The physics of charge inversion in chemical and biological systems, Rev. Modern Phys. 74 (2) (2002) 329.
- [9] Y. Marcus, G. Hefter, Ion pairing, Chem. Rev. 106 (11) (2006) 4585–4621.
- [10] M.R. Tant, K.A. Mauritz, G.L. Wilkes, Ionomers: Synthesis, Structure, Properties and Applications, Springer Science & Business Media, 2012.
- [11] V. Freger, Ion partitioning and permeation in charged low- T^* membranes, Adv. Colloid Interface Sci. 277 (2020) 102107.
- [12] L. Rubatat, G. Gebel, O. Diat, Fibrillar structure of nafion: Matching Fourier and real space studies of corresponding films and solutions, Macromolecules 37 (20) (2004) 7772–7783.
- [13] K.-D. Kreuer, G. Portale, A critical revision of the nano-morphology of proton conducting ionomers and polyelectrolytes for fuel cell applications, Adv. Funct. Mater. 23 (43) (2013) 5390–5397.
- [14] V. Kolev, V. Freger, Molecular dynamics investigation of ion sorption and permeation in desalination membranes, J. Phys. Chem. B 119 (44) (2015) 14168–14179.
- [15] M. Kanduć, W.K. Kim, R. Roa, J. Dzubiella, Aqueous nanoclusters govern ion partitioning in dense polymer membranes, ACS Nano 13 (10) (2019) 11224–11234.
- [16] J. Wu, J. He, J.A. Quezada-Renteria, J. Le, K. Au, K. Guo, M. Xiao, X. Wang, D. Dlamini, H. Fan, et al., Polyamide reverse osmosis membrane compaction and relaxation: Mechanisms and implications for desalination performance, J. Membr. Sci. 706 (2024) 122893.
- [17] M. Heyde, C. Peters, J. Anderson, Factors influencing reverse osmosis rejection of inorganic solutes from aqueous solution, J. Colloid Interface Sci. 50 (3) (1975) 467–487.
- [18] M. Heyde, J. Anderson, Ion sorption by cellulose acetate membranes from binary salt solutions, J. Phys. Chem. 79 (16) (1975) 1659–1664.
- [19] E. Glueckauf, The distribution of electrolytes between cellulose acetate membranes and aqueous solutions, Desalination 18 (2) (1976) 155–172.
- [20] S. Mafe, P. Ramirez, A. Tanioka, J. Pellicer, Model for counterion-membrane-fixed ion pairing and donnan equilibrium in charged membranes, J. Phys. Chem. B 101 (10) (1997) 1851–1856.
- [21] H. Matsumoto, R. Yamamoto, A. Tanioka, Membrane potential across low-water-content charged membranes: Effect of ion pairing, J. Phys. Chem. B 109 (29) (2005) 14130–14136.
- [22] P. Magnico, Ion transport dependence on the ion pairing/solvation competition in cation-exchange membranes, J. Membr. Sci. 483 (2015) 112–127.
- [23] E. Kimani, A. Kemperman, W. Van Der Meer, P. Biesheuvel, Multicomponent mass transport modeling of water desalination by reverse osmosis including ion pair formation, J. Chem. Phys. 154 (12) (2021).
- [24] Y.S. Oren, O. Nir, V. Freger, Analyzing ion uptake in ion-exchange membranes using ion association model, J. Membr. Sci. 690 (2024) 122202.
- [25] M. Stolov, V. Freger, Membrane charge weakly affects ion transport in reverse osmosis, Environ. Sci. Technol. Lett. 7 (6) (2020) 440–445.
- [26] B. Blankert, F.D. Martinez, R. Das, T. Altmann, J.S. Vrouwenvelder, C. Picoreanu, On the discrepancy between nominal and effective charge density of polyamide reverse osmosis membranes, Desalination 578 (2024) 117456.
- [27] Y. Levin, M.E. Fisher, Criticality in the hard-sphere ionic fluid, Phys. A 225 (2) (1996) 164–220.
- [28] G. Orkoulas, M.E. Fisher, A.Z. Panagiotopoulos, Precise simulation of criticality in asymmetric fluids, Phys. Rev. E 63 (5) (2001) 051507.
- [29] A.E. Yaroshchuk, Dielectric exclusion of ions from membranes, Adv. Colloid Interface Sci. 85 (2–3) (2000) 193–230.
- [30] A. Szymczyk, N. Fatin-Rouge, P. Fievet, C. Ramseier, A. Vidonne, Identification of dielectric effects in nanofiltration of metallic salts, J. Membr. Sci. 287 (1) (2007) 102–110.
- [31] V. Freger, Dielectric exclusion, an éminence grise, Adv. Colloid Interface Sci. (2023) 102972.
- [32] J.-N. Aqua, S. Banerjee, M.E. Fisher, Criticality in charge-asymmetric hard-sphere ionic fluids, Phys. Rev. E— Stat. Nonlinear, Soft Matter Phys. 72 (4) (2005) 041501.
- [33] G.M. Geise, D.R. Paul, B.D. Freeman, Fundamental water and salt transport properties of polymeric materials, Prog. Polym. Sci. 39 (1) (2014) 1–42.
- [34] S. Bason, V. Freger, Phenomenological analysis of transport of mono- and divalent ions in nanofiltration, J. Membr. Sci. 360 (1–2) (2010) 389–396.
- [35] A.E. Yaroshchuk, Non-steric mechanisms of nanofiltration: superposition of donnan and dielectric exclusion, Sep. Purif. Technol. 22 (2001) 143–158.
- [36] N. Fridman-Bishop, K.A. Tankus, V. Freger, Permeation mechanism and interplay between ions in nanofiltration, J. Membr. Sci. 548 (2018) 449–458.
- [37] L. Bruni, S. Bandini, Studies on the role of site-binding and competitive adsorption in determining the charge of nanofiltration membranes, Desalination 241 (1–3) (2009) 315–330.
- [38] K.M. Beers, D.T. Hallinan Jr., X. Wang, J.A. Pople, N.P. Balsara, Counterion condensation in Nafion, Macromolecules 44 (22) (2011) 8866–8870.
- [39] A. Münchinger, K.-D. Kreuer, Selective ion transport through hydrated cation and anion exchange membranes I. The effect of specific interactions, J. Membr. Sci. 592 (2019) 117372.
- [40] R. Sujanani, O. Nordness, A. Miranda, L.E. Katz, J.F. Brennecke, B.D. Freeman, Accounting for ion pairing effects on sulfate salt sorption in cation exchange membranes, J. Phys. Chem. B 127 (8) (2023) 1842–1855.
- [41] S.M. Bannon, G.M. Geise, Application of the Born model to describe salt partitioning in hydrated polymers, ACS Macro Lett. 13 (5) (2024) 515–520.
- [42] V. Batygin, I. Toptygin, D.T. Haar, Problems in Electrodynamics, 2nd ed., Academic Press, London, 1978.
- [43] V. Neklyudov, V. Freger, Ion coupling, bonding, and transfer in narrow carbon nanotubes, Small (2024) 2402327.
- [44] C. Böttcher, Theory of Electric Polarization, second ed., Elsevier, Amsterdam, 1973, pp. 129–157.
- [45] V. Freger, G.Z. Ramon, Polyamide desalination membranes: Formation, structure, and properties, Prog. Polym. Sci. 122 (2021) 101451.
- [46] H. Fan, Y. Huang, I.H. Billinge, S.M. Bannon, G.M. Geise, N.Y. Yip, Counterion mobility in ion-exchange membranes: spatial effect and valency-dependent electrostatic interaction, ACS ES&T Eng. 2 (7) (2022) 1274–1286.

- [47] G.S. Manning, Binary diffusion and bulk flow through a potential-energy profile: A kinetic basis for the thermodynamic equations of flow through membranes, *J. Chem. Phys.* 49 (6) (1968) 2668–2675.
- [48] G.S. Manning, Limiting laws and counterion condensation in polyelectrolyte solutions I. Colligative properties, *J. Chem. Phys.* 51 (3) (1969) 924–933.
- [49] K.D. Kreuer, On the development of proton conducting polymer membranes for hydrogen and methanol fuel cells, *J. Membr. Sci.* 185 (1) (2001) 29–39.
- [50] X. Wu, X. Wang, G. He, J. Benziger, Differences in water sorption and proton conductivity between nafion and SPEEK, *J. Polym. Sci. Part B: Polym. Phys.* 49 (20) (2011) 1437–1445.
- [51] K. Chang, G.M. Geise, Dielectric permittivity properties of hydrated polymers: measurement and connection to ion transport properties, *Ind. Eng. Chem. Res.* 59 (12) (2019) 5205–5217.